```
rootdir: C:\Users\SALAS\Documents\taller-calidad\taller-calidad
plugins: cov-7.1.0
collected 78 items

tests/test_integracion.py::test_pipeline_carga_clientes PASSED                                                    [  1%]
tests/test_integracion.py::test_pipeline_devuelve_cantidad_de_filas_procesadas PASSED                             [  2%]
tests/test_integracion.py::test_pipeline_es_idempotente PASSED                                                    [  3%]
tests/test_integracion.py::test_pipeline_reemplaza_el_email_si_el_id_ya_existe PASSED                             [  5%]
tests/test_integracion.py::test_pipeline_con_email_invalido_es_atomico PASSED                                     [  6%]
tests/test_integracion.py::test_pipeline_con_email_vacio_lanza_error PASSED                                       [  7%]
tests/test_integracion.py::test_pipeline_csv_inexistente PASSED                                                   [  8%]
tests/test_integracion.py::test_pipeline_id_no_numerico_lanza_error PASSED                                        [ 10%]
tests/test_reportes.py::test_obtener_password_lee_la_variable_de_entorno PASSED                                   [ 11%]
tests/test_reportes.py::test_obtener_password_sin_variable_falla_con_mensaje_claro PASSED                         [ 12%]
tests/test_reportes.py::test_buscar_cliente_encuentra_coincidencia_exacta PASSED                                  [ 14%]
tests/test_reportes.py::test_buscar_cliente_sin_coincidencias PASSED                                              [ 15%]
tests/test_reportes.py::test_buscar_cliente_resiste_inyeccion_sql[' OR '1'='1] PASSED                             [ 16%]
tests/test_reportes.py::test_buscar_cliente_resiste_inyeccion_sql[x' OR 1=1 --] PASSED                            [ 17%]
tests/test_reportes.py::test_buscar_cliente_resiste_inyeccion_sql['; DROP TABLE clientes; --] PASSED              [ 19%]
tests/test_reportes.py::test_evaluar_regla_validas[monto > 100-fila0-True] PASSED                                 [ 20%]
tests/test_reportes.py::test_evaluar_regla_validas[monto > 100-fila1-False] PASSED                                [ 21%]
tests/test_reportes.py::test_evaluar_regla_validas[monto >= 100 and id < 5-fila2-True] PASSED                     [ 23%]
tests/test_reportes.py::test_evaluar_regla_validas[monto >= 100 and id < 5-fila3-False] PASSED                    [ 24%]
tests/test_reportes.py::test_evaluar_regla_validas[monto < 10 or id == 1-fila4-True] PASSED                       [ 25%]
tests/test_reportes.py::test_evaluar_regla_validas[not (monto > 10)-fila5-True] PASSED                            [ 26%]
tests/test_reportes.py::test_evaluar_regla_validas[0 < monto <= 100-fila6-True] PASSED                            [ 28%]
tests/test_reportes.py::test_evaluar_regla_validas[0 < monto <= 100-fila7-False] PASSED                           [ 29%]
tests/test_reportes.py::test_evaluar_regla_validas[monto * 2 - 1 == 9-fila8-True] PASSED                          [ 30%]
tests/test_reportes.py::test_evaluar_regla_validas[-monto < 0-fila9-True] PASSED                                  [ 32%]
tests/test_reportes.py::test_evaluar_regla_validas[monto / 2 == 2.5-fila10-True] PASSED                           [ 33%]
tests/test_reportes.py::test_evaluar_regla_validas[estado == 'activo'-fila11-True] PASSED                         [ 34%]
tests/test_reportes.py::test_evaluar_regla_validas[estado != 'activo'-fila12-False] PASSED                        [ 35%]
tests/test_reportes.py::test_evaluar_regla_rechaza_construcciones_peligrosas[__import__('os').system('echo hack')] PASSED [ 37%]
tests/test_reportes.py::test_evaluar_regla_rechaza_construcciones_peligrosas[monto.real > 1] PASSED               [ 38%]
tests/test_reportes.py::test_evaluar_regla_rechaza_construcciones_peligrosas[[x for x in range(3)]] PASSED        [ 39%]
tests/test_reportes.py::test_evaluar_regla_rechaza_construcciones_peligrosas[monto ** 2 > 1] PASSED               [ 41%]
tests/test_reportes.py::test_evaluar_regla_rechaza_construcciones_peligrosas[monto in (1, 2)] PASSED              [ 42%]
tests/test_reportes.py::test_evaluar_regla_rechaza_construcciones_peligrosas[lambda: 1] PASSED                    [ 43%]
tests/test_reportes.py::test_evaluar_regla_campo_desconocido PASSED                                               [ 44%]
tests/test_reportes.py::test_evaluar_regla_sintaxis_invalida PASSED                                               [ 46%]
tests/test_reportes.py::test_total_suma_la_segunda_columna PASSED                                                 [ 47%]
tests/test_reportes.py::test_total_de_lista_vacia_es_cero PASSED                                                  [ 48%]
tests/test_reportes.py::test_total_con_formato_invalido_lanza_value_error[filas0] PASSED                          [ 50%]
tests/test_reportes.py::test_total_con_formato_invalido_lanza_value_error[filas1] PASSED                          [ 51%]
tests/test_reportes.py::test_total_con_formato_invalido_lanza_value_error[None] PASSED                            [ 52%]
tests/test_transform.py::test_normalizar_email_valido[  Ana@Mail.COM -ana@mail.com] PASSED                        [ 53%]
tests/test_transform.py::test_normalizar_email_valido[luis.p+datos@uni.edu.co-luis.p+datos@uni.edu.co] PASSED     [ 55%]
tests/test_transform.py::test_normalizar_email_valido[ANA@MAIL.COM-ana@mail.com] PASSED                           [ 56%]
tests/test_transform.py::test_normalizar_email_valido[\tpedro@mail.com\n-pedro@mail.com] PASSED                   [ 57%]
tests/test_transform.py::test_normalizar_email_valido[a_b-c@sub-dom.mail.com-a_b-c@sub-dom.mail.com] PASSED       [ 58%]
tests/test_transform.py::test_normalizar_email_invalido[] PASSED                                                  [ 60%]
tests/test_transform.py::test_normalizar_email_invalido[   ] PASSED                                               [ 61%]
tests/test_transform.py::test_normalizar_email_invalido[sin-arroba] PASSED                                        [ 62%]
tests/test_transform.py::test_normalizar_email_invalido[None] PASSED                                              [ 64%]
tests/test_transform.py::test_normalizar_email_invalido[123] PASSED                                               [ 65%]
tests/test_transform.py::test_normalizar_email_invalido[@mail.com] PASSED                                         [ 66%]
tests/test_transform.py::test_normalizar_email_invalido[ana@] PASSED                                              [ 67%]
tests/test_transform.py::test_normalizar_email_invalido[ana@mail] PASSED                                          [ 69%]
tests/test_transform.py::test_normalizar_email_invalido[ana mail@x.com] PASSED                                    [ 70%]
tests/test_transform.py::test_normalizar_email_invalido[ana@@mail.com] PASSED                                     [ 71%]
tests/test_transform.py::test_normalizar_email_vacio_mensaje[] PASSED                                             [ 73%]
tests/test_transform.py::test_normalizar_email_vacio_mensaje[   ] PASSED                                          [ 74%]
tests/test_transform.py::test_normalizar_email_vacio_mensaje[None] PASSED                                         [ 75%]
tests/test_transform.py::test_normalizar_email_vacio_mensaje[42] PASSED                                           [ 76%]
tests/test_transform.py::test_normalizar_email_invalido_mensaje_incluye_original PASSED                           [ 78%]
tests/test_transform.py::test_calcular_total_sin_descuento PASSED                                                 [ 79%]
tests/test_transform.py::test_calcular_total_con_descuento PASSED                                                 [ 80%]
tests/test_transform.py::test_calcular_total_descuento_limites[0-200.0] PASSED                                    [ 82%]
tests/test_transform.py::test_calcular_total_descuento_limites[1-0.0] PASSED                                      [ 83%]
tests/test_transform.py::test_calcular_total_cero[0-5] PASSED                                                     [ 84%]
tests/test_transform.py::test_calcular_total_cero[50-0] PASSED                                                    [ 85%]
tests/test_transform.py::test_calcular_total_cero[0-0] PASSED                                                     [ 87%]
tests/test_transform.py::test_calcular_total_negativos[-1-1] PASSED                                               [ 88%]
tests/test_transform.py::test_calcular_total_negativos[1--1] PASSED                                               [ 89%]
tests/test_transform.py::test_calcular_total_negativos[-5--5] PASSED                                              [ 91%]
tests/test_transform.py::test_calcular_total_descuento_fuera_de_rango[-0.01] PASSED                               [ 92%]
tests/test_transform.py::test_calcular_total_descuento_fuera_de_rango[1.01] PASSED                                [ 93%]
tests/test_transform.py::test_calcular_total_descuento_fuera_de_rango[2] PASSED                                   [ 94%]
tests/test_transform.py::test_calcular_total_descuento_fuera_de_rango[-1] PASSED                                  [ 96%]
tests/test_transform.py::test_calcular_total_redondeo_a_dos_decimales[19.99-3-0.1-53.97] PASSED                   [ 97%]
tests/test_transform.py::test_calcular_total_redondeo_a_dos_decimales[0.1-3-0.0-0.3] PASSED                       [ 98%]
tests/test_transform.py::test_calcular_total_redondeo_a_dos_decimales[9.99-1-0.5-5.0] PASSED                      [100%]

---------- coverage: platform win32, python 3.12.10-final-0 ----------

Name               Stmts   Miss Branch BrPart  Cover   Missing
--------------------------------------------------------------
etl\__init__.py        0      0      0      0   100%
etl\pipeline.py        9      0      0      0   100%
etl\reportes.py       51      0     20      0   100%
etl\transform.py      13      0      6      0   100%
--------------------------------------------------------------
TOTAL                 73      0     26      0   100%
PS C:\Users\SALAS\Documents\taller-calidad\taller-calidad>
```